NIGRAH Threat Index & Mathematical Scoring Framework (TISF v1)

Why This Document Matters

This is the document that transforms NIGRAH from a classification system into an intelligence system.

Most fraud detection systems answer:

Is this content suspicious?

NIGRAH answers:

How dangerous is this threat?

How quickly is it spreading?

How much damage has it historically caused?

Which regions are most exposed?

How confident are we in the prediction?

This is the intelligence layer.

The Core Equation

Every detected threat receives a:

Threat Index (TI)

Scale:

```
0 - 100
```

Where:

```
0-20    Low
21-40   Moderate
41-60   Elevated
61-80   High
81-100  Critical
```

Threat Index Formula

Initial framework:

$$TI = 0.30(SPS) + 0.25(FHS) + 0.15(GRS) + 0.15(RIS) + 0.15(CS)$$

Where:

Variable	Meaning
SPS	Signal Pattern Score
FHS	Family Harm Score
GRS	Growth Rate Score
RIS	Regional Impact Score
CS	Confidence Score

Component 1

Signal Pattern Score (SPS)

Measures:

How dangerous are the manipulation signals present?

Formula:

```
SPS

=

Σ(Signal Weight × Signal Confidence)

÷

Maximum Possible Weight
```

Example

Detected:

```
Guaranteed Return
Weight = 10

Urgency
Weight = 8

Fake Authority
Weight = 9

Social Proof
Weight = 8
```

Confidence

```
Guaranteed Return = 0.95

Urgency = 0.90

Authority = 0.88

Social Proof = 0.80
```

Calculation

```
(10×0.95)
+
(8×0.90)
+
(9×0.88)
+
(8×0.80)
=
31.02
```

Normalized:

```
SPS = 82
```

Interpretation

This content strongly resembles known threat signals.

Component 2

Family Harm Score (FHS)

Measures:

Historically, how harmful is this scam family?

Formula

```
FHS
```

=

```
Severity
```

×

```
Victim Volume
```

×

```
Financial Damage
```

Normalized to:

```
0 – 100
```

Example

Fake Trading App Family

```
Severity = 95

Victim Volume = 80

Financial Loss = 90
```

Result:

```
FHS = 88
```

This means:

Historically this family causes significant damage.

Component 3

Growth Rate Score (GRS)

Measures:

Is this scam family expanding?

Formula

```
GRS
=
(Current Mentions)
/
(Historical Baseline)
```

Example

Previous Month

```
100 reports
```

Current Month

```
260 reports
```

Growth

```
260 / 100
=
2.6x
```

GRS:

```
92
```

Meaning

Threat is accelerating.

Component 4

Regional Impact Score (RIS)

Measures:

Is this scam especially active in a region?

Example

Region:

```
Uttar Pradesh
```

Observed:

```
Telegram Fraud = High

Fake Advisor Fraud = Very High
```

Regional Weight:

```
UP = 85
```

NIGRAH could eventually produce:

```
State-wise Threat Maps
```

Example:

State	Risk
UP	85
Bihar	82
MP	77
Rajasthan	74

This is a differentiator.

Most systems ignore geography.

Component 5

Confidence Score (CS)

Measures:

How confident are we?

This prevents overclaiming.

Sources contribute different confidence.

Example:

Source	Confidence
SEBI Order	100
RBI Alert	100
Court Judgment	95
News Article	80
Public Complaint	65
Social Media	40

Formula

```
CS
=
Weighted Source Reliability
```

Example

If a threat is supported by:

```
SEBI
+
```

```
News

+

Court
```

Result:

```
CS = 91
```

The NIGRAH Confidence Principle

Instead of saying:

```
This is a scam.
```

We say:

```
This content has

87% similarity

to a known

Critical-Risk Scam Family

with

91% confidence.
```

This is much safer.

Much more explainable.

Regional Threat Intelligence

This may become one of the strongest features.

Every scam family gets:

```
Regional Penetration Score
```

Example

```
{
  "family": "Telegram Signal Fraud",
  "UP": 89,
  "Bihar": 82,
  "MP": 73,
  "Maharashtra": 68
}
```

Now NIGRAH becomes:

```
National Threat Intelligence Infrastructure
```

instead of:

```
Screenshot Scanner
```

Emerging Threat Index (ETI)

This is potentially patent-worthy.

Most systems identify known scams.

NIGRAH should identify:

Emerging Scam Families

Formula:

```
ETI
=
Signal Novelty
```

```
×
```

```
Growth Velocity
```

```
×
```

```
Pattern Similarity
```

Example

New scam appears.

No existing family match.

But:

```
Authority = High
```

```
Greed = High
```

```
Urgency = High
```

```
Identity Manipulation = High
```

Similarity:

```
74%
```

to known families.

Growth:

```
300%
```

month-over-month.

Result:

```
ETI = 91
```

Meaning:

This may be an emerging scam ecosystem.

National Scam Severity Index (NSSI)

This is the macro layer.

Every family contributes:

```
Threat Index

×

Victim Count

×

Financial Harm
```

Result:

National ranking.

Example:

Family	NSSI
Fake Trading App	96
Telegram Signal Fraud	92
Fake Advisor Fraud	90
Deepfake Advisor	85
Ponzi Variants	83

This becomes a regulator-grade dashboard.

The NIGRAH Intelligence Stack

At this stage we have:

Layer 1

Threat Signals

25 Signals

↓

Layer 2

Manipulation Patterns

10 Patterns

↓

Layer 3

Scam Families

30 Families

↓

Layer 4

Threat Index Engine

(TI, FHS, GRS, RIS, CS)

↓

Layer 5

Emerging Threat Engine

(ETI)

↓

Layer 6

National Threat Dashboard

(NSSI)